SECURE FEEDER OPERATION
IN
RADIAL DISTRIBUTION SYSTEM

A Project Report Submitted

In Partial Fulfillment of the Requirements

for the awards of Degree

of

BACHELOR OF TECHNOLOGY
in
ELECTRICAL ENGINEERING

by

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UNDERTAKING

We declare that the work presented in this report titled **“Secure Feeder Operation in Radial Distribution System”** submitted to the **Department of Electrical Engineering, Rajkiya Engineering College, Ambedkar Nagar,** for the award of the **Bachelor of Technology** degree**,** is our original work. We have not plagiarized or submitted the same work for the award of any other degree. In case this undertaking is found incorrect, we accept that our degree may be unconditionally withdrawn.

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This is to certify that **Mr. Praveen Gautam (1907370200036), Mr. Pushkar Kumar (1907370200038) and Mr. Rajneesh Kumar (1907370200040)** has carried out the project work presented in this report entitled **"Secure Feeder Operation in Radial Distribution System"** for the award of **Bachelor of Technology** in **Electrical Engineering** from **Dr. A. P. J. Abdul Kalam Technical University Lucknow** under my supervision during the academic session 2022-23.

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PREFACE

Security of radial distribution feeder is requirement of the current scenario. It may be achieved by secure operation of switchgear directly connected to the feeder. To avoid various critical electrical accidents during maintenance of the feeder due to lack of communication and coordination between the maintenance staff and electric substation staff, the secure operation of circuit breaker is necessary. This paper presents a password base secured circuit breaker operation methodology. To avoid electrical accidents this project is designed, in which there is a password-based circuit breaker system that can only be access by specified password to control the circuit breaker by authorized person only. The system is fully controlled by the 8-bit Microcontroller from 8051 family which has an 8 KB of RAM for the program memory. A matrix keypad is used to enter the password and Relay driver IC is used to switch ON/OFF the loads through relays. A LCD will be used to see the written password.

CONTENTS

List of Figures

LIST OF ABBREVIATIONS

DC	Direct Current
AC	Alternating Current
SPDT	Single Pole Double Through
LED	Light Emitting Diode
LCD	Liquid Crystal Display
NPN Transistor	Negative-Positive-Negative Transistor
PNP Transistor	Positive-Negative-Positive Transistor
V.R.	Voltage Regulator

CHAPTER 1

INTRODUCTION

Electrical accidents of the line man are increasing while repairing the electrical lines due to lack of communication between the electrical substation and maintenance staff. The main objective of this project is secure operation to distribution feeder as well as provide safety to the electrical lineman with the help of password-based circuit breaker that can only be access by specified password to control the circuit breaker by authorized person only. This system is used to control to turn ON/OFF the line with the authorized person only. Now if there is any fault in electrical line then authorized person will switch off the power supply by entering the correct password and comfortably clearing the fault, after that authorized person turn on the power supply by entering the correct password only. This system is fully controlled by microcontroller from the 8051 family. Hex keypad is interfaced with microcontroller to enter the password, If the entered password is correct then the microcontroller performs further operations. 16x2 LCD is also connected to microcontroller to display the message. SPDT relays are also present here to sense the signal from microcontroller and then perform open/close the circuit.

The following are the key features:

- In this prototype we can easily break the line by entering password.
- This proposed system provide a solution, which can ensure the safety maintenance staff.
- This system has an arrangement such that a password is required to operate the circuit breaker (ON/OFF).
- Authorized person can turn on/off the supply

1.1. MOTIVATION

The major problem in the power system is the electrical accidents while repairing the electrical lines due to the lack of communication between the electrical substation andmaintenance staff. This project gives a solution to this problem to ensure line man safety. So we have made this project to ensure the safety of the lineman. In this project we have made a password based circuit breaker that ON/OFF the supply when correct password is written. So when a lineman goes on site to work he can cut the power supply using password, even if the staff and lineman have miscommunication about power supply the staff at office can’t turn ON the power supply without the password that is only known to lineman only. This way lineman is protected and when he finishes his work at site, he can turn ON the supply using password. Hence there will be no accidents. Our motivation for this project is that we are saving lives through this project.

1.2. LITERATURE SURVEY

[1] Athira P Nair, Josephin J, Anjana A S, Athira C P, Sebin J Olickal "ELECTRIC LINE MAN SAFETY SYSTEM WITH OTP BASED CIRCUIT BREAKER". Volume: 04 Special Issue: 03 | Apr-2015

Contribution of the paper: The electric line man safety system is designed to control a circuit breaker with help of a password only. OTP generation and OTP verification are the major tasks involved in this system. OTP generation is the main attraction of this project. It provides a new approach to the security of the lineman and completely eliminates the accidents to the lineman due to electric shock during the electric line repair.

Scope: Using wireless communication this system can be operated from other areas besides the substation such as on the transformer. The SCADA is a system used in the communication channels to help easy troubleshoot to locate the fault location directly and the line man can easily rectify it.

[2] Pramod M. Murari, Mahabal V. Kinnerkar, Prashant S. Koppa, Vishal S. Kamble, Rashmitha R. Mendan"Electric Line Man Safety with Password Based Circuit Breaker and Intimation of HT Wire Sag using GSM" Volume 2, Issue 7 | July 2017

Contribution of the paper: It can work on a single given known password Whenever the HT wire sag is higher than the predetermined range/value then ultrasonic sensor sends a message to the substation through GSM and intimates the operator about the trouble occurred. PIR sensor provides safety to human/animal by sensing their presence and alerting through a buzzer. It can be concluded that the proposed system can be used as an effective application in the present working system and provides safety to lineman and corrective measures can be taken after HT wire sag intimation.

Scope: Instead of GSM 300 we can use GSM 900 which can be connected to IOT (internet of things). By using IOT we can operate the relays from any area as we can directly connect to the server. Wireless ultrasonic and PIR sensors can be also used. We can use SCADA system, to help easy trouble shoot, to identify the fault location directly and line man can easily rectify it. we can also use EPROMS that can be interfaced to system so the circuit breaker cannot only operate from the substation, but also from other location through wireless communication.

[3] S.P. Achrekar, Shubham Jadhav, Praful Gurav, Makrand Rawool, Kaushal Manjrekar "OTP Circuit Breaker for Lineman Safety and Maintenance", Volume:05 Issue 01|2018.

Contribution of the paper: This project control system is a system that accesses only specified password to control the circuit breaker. Here, there is also a provision of changing the password. The system is fully controlled by the 8-bit microcontroller from 8051 family which has an 8KB of ROM for the program memory. A matrix keypad is interfaced to the microcontroller to enter the password, while a relay driver IC is used to switch ON / OFF the loads through relays. The complete circuit is built with on board power supply. The power supply consists of a step-down transformer 230/12V, which steps down the voltage to 12V AC. This is converted to DC using a Bridge rectifier. The ripples are removed using a capacitive filter and it is then regulated to +5V using a voltage regulator which is required for the operation of the microcontroller and other components.

Scope: The SCADA is a system used in the communication channels to help easy troubleshoot to locate the fault location directly and the line man can easily rectify it.

[4] Rakesh Narvey, Rahul Sagwan, Abhishek Katroliya, Vipin Jain, Hari Singh Sikarwar, Krishnakant "Password Based Circuit Breaker Using Aurdino" , IJAEM Volume:02 Issue 01|15-07-2020

Contribution of the paper: To conclude with the project it's that the proposed system is a simple design and low budget economical system. The safety and the protection of lineman is our priority. Hence, we tried to develop a system that can ensure it with full accuracy. Finally, the aim of the project i.e., to avoid the fatal accident for the lineman.

Scope: We can placed sensor for each and every lineman to detect the fault and automatic send fault SMS to lineman for repair of line.

[5] Mr. Dilip D. Gaikawad, Mr. Akshay P. Awari, Mr. Yugal M. Mahajan, Mr. Yuvraj G. Kharde, Mr. Rushikesh R. Deshmukh "Password Based Circuit Breaker Using GSM Module for Wireman Safety", IJRASET Volume:07 Issue:06|06-2019.

Contribution of the paper: By using the implementation, we can say that the improvement in advance technology we can clear the fault easily and it is safer to use for quick response. Also, it is easy to utilize and simple in programming that will enables us to control various system easily.

Scope: By implementation of PLC and SCADA technology one can easily monitor of the location healthy and faulty zone and keep it isolated until it repaired.

[6] Vaibhavi Purohit, Rohit Kamble, Vivek Bhui, Suraj Patil, Sunil Salaskar, Om Chougule "Protection foe Electric Lineman Using Mobile Application with a Code Based Circuit Breaker", IJARIIE Volume:07 Issue 02|2021.

Contribution of the paper: Serious electrical injuries to lineman are on the rise during an line repair because of loss of coordination and interaction between the maintenance workers and electric substation staff, The purpose of this paper is to provide protection of an electric man, the project is programmed to monitor the circuit breaker using a password. A circuit breaker can be enabled with a single password and password can be modified when needed, it eliminates the possibility of password theft. It is successful in ensuring the protection of the staff.

Scope: This proposed system offers a solution for ensuring the safety of maintenance personnel, such as lineman. Since this device is set up in such a way that a password is required to operate the circuit breaker (ON/OFF), the lineman will be in charge of the turning on/off line.

[7] B.Sai Kumar, G. Vijay Kumar, G. Sai Krishna, J.V.S Nanda. M.V.S.S.R. Krishnakanth, R. Sai Sunandh "Password Based Circuit Breaker Using Aurdino", Pramana Research Journal Volume:10 Issue04|2020.

Contribution of the paper: In the current scenario, accidental death of lineman is often read and evidenced, a safety measure to safeguard the operator is found very necessary looking into the present working style. The electric lineman safety system is designed to control the control panel doors and circuit breaker by using a password for the safety. It can work on a single password, password can be changed by authorized person, no other person can reclose the breaker. It is effective in providing safety to the working staff.

Scope: The proposed system is designed to ensure the safety of lineman with the help of password-based circuit breaker using aurdino.

[8] S.P. Manikanta, Srinivasarao.T, P.A. Lovina, "Passoword Based Circuit Breaker", JES Volume:11 Issue 06|June 2020.

Contribution of the paper: Nowadays electric accidents to the linemen are increasing while repairing the electrical line. This project gives a solution to ensure the safety of linemen. In this proposed project work, the control (ON/OFF) of the electric line lies with the line man. The concept is designed such that maintenance staff, or the line man must enter password to switch (ON/OFF) the electric line.

Scope: This system provides a solution which can ensure that only the lineman can control the system and thus there is no possibility of someone else interfering the system.

[9] G. Chandana, L. Sandhya, K. Revanth, Aaqib-Ur-Rehman "Password Based Circuit Breaker Using 8051", IJCRT Volume:09 Issue 01|Jan 2021.

Contribution of the paper: When operated manually we see fatal electrical accidents to the lineman are increasing during the electrical line repair due to the lack of communication and coordination between the maintenance staff and electric substation staff. In order to avoid such accidents, the breaker can be so designed such that only an authorized person can operate with a password only. This ensures security of the worker because no one can turn ON the line without password.

Scope: In order to avoid electrical accidents, the breaker can be so designed such that only authorized person can operate it with a password.

[10] C. Pearline Kamalini, A. Kokila, S. Jesimabanu, V. Jaylakshmi "Password Based Circuit Breaker Control to Ensure the Electric Lineman Safety and Load Sharing", IJERT Volume:05 Issue 13|2017.

Contribution of the paper: The major problem in the power system is the electrical accidents while repairing the electric line due to miscommunication between the maintenance staff and electric substation staff. This paper gives a solution to this problem to ensure lineman safety, also the load distribution system has been proposed in which sharing of load is done between village side and city side.

Scope: This project provides the safety to electrical workers without hesitation with the help of password-based circuit breaker. It can work on given known password and reduced load demand in the distribution side.

CHAPTER 2

Hardware Components

2.1. Microcontroller (AT89S52):

The AT89S52 is a low-power, high-performance CMOS 8-bit microcontroller with 8Kbytes of in-system programmable Flash memory. The device is manufactured. Using Atmel's high-density non-volatile memory technology and is compatible with the industry-standard 80C51 micro controller. The on-chip Flash allows the program memory to be reprogrammed in-system or by a conventional non-volatile memory programmer. By combining a versatile 8-bit CPU with in-system programmable flash one monolithic chip; the Atmel AT89S52 is a powerful micro controller, which provides a highly flexible and cost-effective solution to many embedded control applications. Features of AT89S52 are listed below.

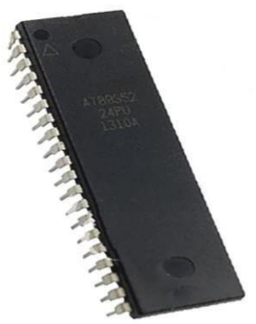

Figure 1 Microcontroller (AT89S52)

- Compatible with MCS-51 Products
- 8K Bytes of In-System Programmable (ISP) Flash Memory
- Endurance: 1000 Write/Erase Cycles
- 4.0V to 5.5V Operating Range
- Fully Static Operation: 0 Hz to 33 MHz
- Three-level Program Memory Lock
- 256bytes Internal RAM
- 32 Programmable I/O Lines
- 16-bit Timer/Counters
- Eight Interrupt Sources
- Full Duplex UART Serial Channel
- Low-power Idle and Power-down Modes
- Interrupt Recovery from Power-down Mode
- Watchdog Timer
- Dual Data Pointer
- Power-off Flag

2.2. HEX Keypad:

HEX keypad is a standard device with 12 keys connected in a 3x4 matrix, giving the characters 0-9. Interfacing of Hex key pad to AT89S52 is essential while designing embedded system projects which require character or numeric input or both. For example projects like digital code lock, numeric calculator etc. Here we are using this to enter numericpassword for turn ON/OFF the circuit breaker. This can be easily interfaced with ant kits Microcontroller Development Board. It is a four pin tactile switch and four mounting holes 3.2mm each.

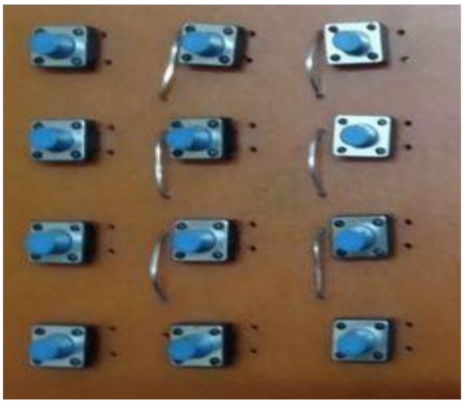

Figure 2 HEX Keypad

2.3. Relay:

A relay is an electrically operated switch. Many relays use an electromagnet to operate a switching mechanism mechanically, but other operating principles are also used. Relays are used where it is necessary to control a circuit by a low-power signal (with complete electrical isolation between control and controlled circuits), or where several circuits must be controlled by one signal. A relay is an electrically operated switch. Current flowing through the coil of the relay creates a magnetic field which attracts a lever and changes the switch contacts. The coil current can be on or off, so relays have two switch positions and most have double throw (changeover) switch contacts as shown in the diagram. In this project we are using Single pole double through (SPDT) relay.

Figure 3 SPDT Relay

2.4. LCD Display:

For ease of interaction with the user, this system uses an electronic display module. Here a 16x2 LCD is used. This means in 2 lines it is possible to display 16 characters per line. A 5x8 pixel matrix is used for display one character. Two registers are associated with an LCD, such as data and command.

These modules are preferred since it is easily programmable. For providing visual assistance to the lineman this module is unavoidable. Here this used to display the password entered by us to ON/OFF the circuit breakers.

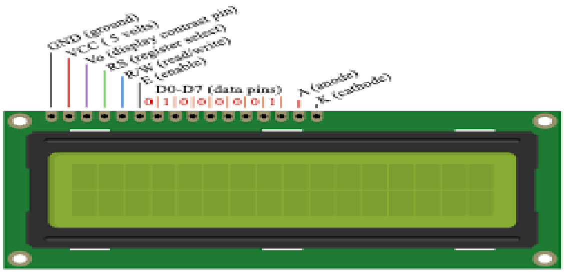

Figure 4 LCD Display

2.5. Step Down Transformer:

A step-down transformer is an electrical device that reduces the voltage of an alternating current (AC) power supply. It consists of a primary winding, a secondary winding, and an iron core. When an AC voltage is applied to the primary winding, it creates a fluctuating magnetic field in the iron core. This magnetic field then induces a voltage in the secondary winding, but at a lower voltage level than the primary winding. Here step-down transformer reduces the 220volts into 12volts.

Figure 5 Step Down Transformer

2.6. Voltage Regulator:

A voltage regulator is designed to automatically maintain a constant voltage level. A voltage regulator may use an electromechanical mechanism, or electronic components. Depending on design, it may be used to regulate one or more voltages. 7805 voltage regulating IC is used to provide the voltage of 5V DC.

Figure 6 Voltage Regulator

2.7. Capacitor:

A capacitor is an electrical device that can store energy in the electric field between a pair of closely-spaced conductors (called 'plates'). When voltage is applied to the capacitor, electric charges of equal magnitude, but opposite polarity, build up on each plate. Capacitors are used in electrical circuits as energy storage devices. They can also be used to differentiate between high-frequency and low frequency signals and this makes them usefulin electronic filters. Capacitors are occasionally referred to as condensers. This is nowconsidered an antiquated term electrolytic capacitor. An electrolytic capacitor is a type of capacitor typically with a larger capacitance per unit volume than other types, making them valuable in relatively high current and Low-frequency electrical circuits. Here we are using 1000 microfarad and 470microfarad 25-volt capacitor.

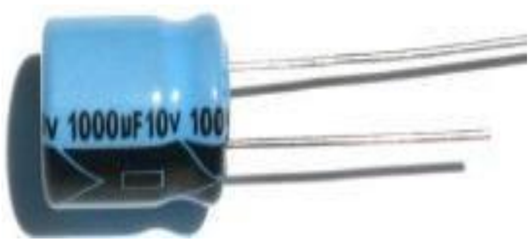

Figure 7 Capacitor

2.8. Crystal Oscillator:

Crystal oscillator is an electronic oscillator circuit that uses mechanical resonance of a vibrating crystal of piezoelectric material to create an electrical signal with a very precise frequency. This frequency is commonly used to keep track of time to provide a stable clock signal for digital integrated circuits, and to stabilize frequencies for radio transmitters and receivers. The most common type of piezoelectric resonator used is thequartz crystal, so oscillator circuits incorporating them became known as crystal oscillator. Here we are using 11.0592MHz crystal oscillator.

Figure 8 Crystal Oscillator

2.9. Resistor:

A resistor is a passive component that implements electrical resistance circuit element. Resistors act to reduce current flow, and, at the same time, act to lower voltagelevels within circuits. Resistors may have fixed resistances or variable crystals tend to two-terminal electrical as a resistances, such as those found in thermistors trimmers, photo resistors and potentiometers. The current through a resistor is in direct proportion the voltage across the resistor's terminals. *R* is the resistance of the conductor in units of ohms (symbol: Ω). The ratio of the voltage applied across a resistor's terminals to the intensity of current in the circuit is called its resistance, and this can be assumed to be a constant (independent ofthe voltage) for ordinary resistors working within their ratings.

Figure 9 Resistor

2.10. Diode:

A diode is an electronic device having a two-terminal unidirectional power supply. A semiconductor diode is the first diode that forms in semiconductor electronic devices, after that there were lots of new innovations takes place. The most common diode used is a semiconductor diode. Diodes can be used as rectifiers, signal limiters, voltage regulators, switches, signal modulators, signal mixers and oscillators. Here we are using 1N4007 diode.

Figure 10 Diode

2.11. PNP Transistor:

A PNP transistor is a bipolar junction transistor constructed by sandwiching an N-type semiconductor between two P-type semiconductors. It is a current controlled device. A PNP transistor has three terminals – a Collector (C), Emitter (E) and Base (B). The PNP transistor behaves like two PN junctions diodes connected back to back. PNP transistors are used to source current, i.e. current flows out of the collector. PNPtransistors are used as switches; they are also used in the amplifying circuits. They are used when we need to turnoff something by push a button. i.e. emergency shutdown. Also used in matched pair circuits to produce continuous power, heavy motors to control current flow, robotic applications. Here we are using BC557 PNP Transistor.

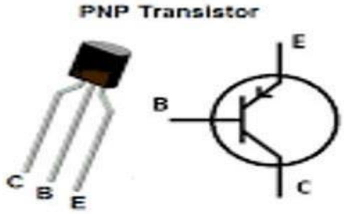

Figure 11 PNP Transistor

2.12. NPN transistor:

An NPN transistor is the most commonly used bipolar junction transistor, and is constructed by sandwiching a P-type semiconductor between two N-type semiconductors. An NPN transistor has three terminals– a collector, emitter and base. The NPN transistor behaves like two PN junctions diodes connected back to back. The NPN transistor amplifies the weak signal enter into the base and produces strong amplify signals at the collector end. In NPN transistor, the direction of movement of an electron is from the emitter to collector region due to which the current constitutes in the transistor. Such type of transistor is mostly used in the circuit because their majority charge carriers are electrons which have high mobility as compared to holes. NPN transistors are used in amplifying circuit applications, Darlington pair circuits for amplifying weak signals, also in applications when we need sinking current, classic amplifier circuits, the same as 'push-pull' amplifier circuits. Here we are using BC547 NPN Transistor.

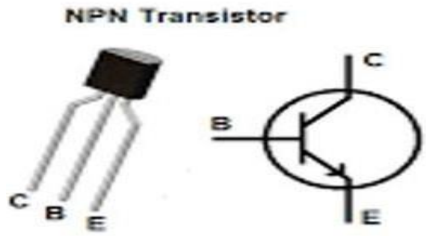

Figure 12 NPN Transistor

2.13. Ceramic Capacitor:

A ceramic capacitor refers to a fixed-value capacitor in which the ceramic material performs the role of a dielectric. Its construction takes place with multiple alternating ceramic layers as well as a metal layer. Furthermore, the metal layer performs the role of electrodes. Ceramic capacitors are mainly used for high stability performances and low-loss devices. These devices provide very accurate results, and also, thecapacitance values of these capacitors are stable with respect to the applied voltage, frequency and temperature.

Figure 13 Ceramic Capacitor

2.14. LED:

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it. When current passes through an LED, the electrons recombine with holes emitting light in the process. LEDs allow the current to flow in the forward direction and block the current in the reverse direction. Light-emitting diodes are heavily doped p-n junctions. Based on the semiconductor material used and the amount of doping, an LED will emit coloured light at a particular spectral wavelength when forward biased. As shown in the figure, an LED is encapsulated with a transparent cover so that emitted light can come out.

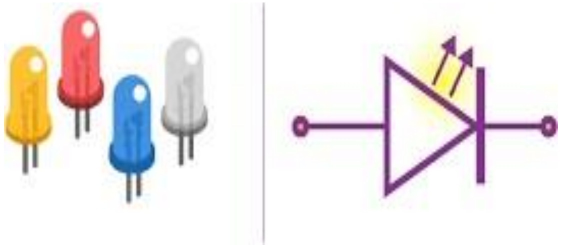

Figure 14 LEDs

Here we are using LEDs to indicate the status of the Loads whether the load is ON or OFF. If the LED is on that means Load is ON condition, if the LED is off that means Load is in off condition.

2.15. Pendant Bulb Holder:

Pendant Holders are the type of bulb holders that areused to hang fancy lights from the wall or ceiling. This type of bulb holder hangs down witha portable wire that is connected to a ceiling rose. It helps to secure a light or lamp directly from the wall or ceiling.

Figure 15 Pendant Bulb Holder

2.16. Electric Bulb:

The electronic bulb is the simplest electrical lamp that was invented for illumination more than a century ago. It was the small and simplest light that brightened the dark space. The electronic bulb is also known as an incandescent lamp, incandescent light globe or incandescent light bulb. Bulb comes in different sizes and light output and operates with a voltage range from 1.5 Volts to about 300 Volts.

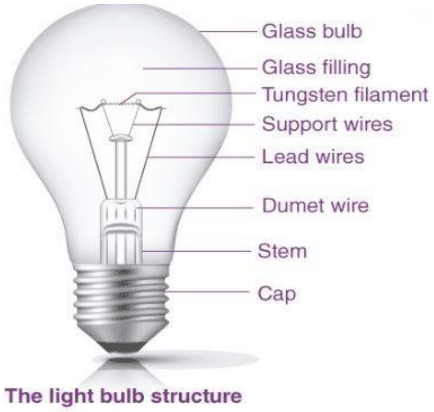

Figure 16 Electric Bulb

2.17. PCB Board:

A printed circuit board is used to mechanically support and electrically connect electronic components using conductive pathways, tracks or signal traces etchedfrom copper sheets laminated onto a non-conductive substrate. It is a medium used inelectrical and electronic engineering to connect electronic components to one another in a controlled manner.

Figure 17 PCB Board

2.18. Electric Socket Board:

A device into which an electric plug can be inserted in order to make a connection in a circuit. Here we are using the electric socket board as Load or feeder distribution.

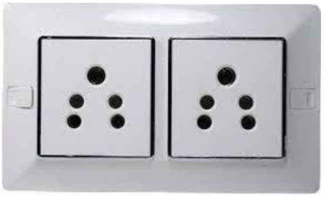

Figure 18 Electric Socket

CHAPTER 3

Problem Formulation

It is observed at the time of fault clearing that electrical accidents occur due the miscommunication between the lineman and the substation staff. This system is designed to provide a solution to ensure the safety of linemen so, this system is designed to ensure the safety of linemen.

3.1. Block Diagram:

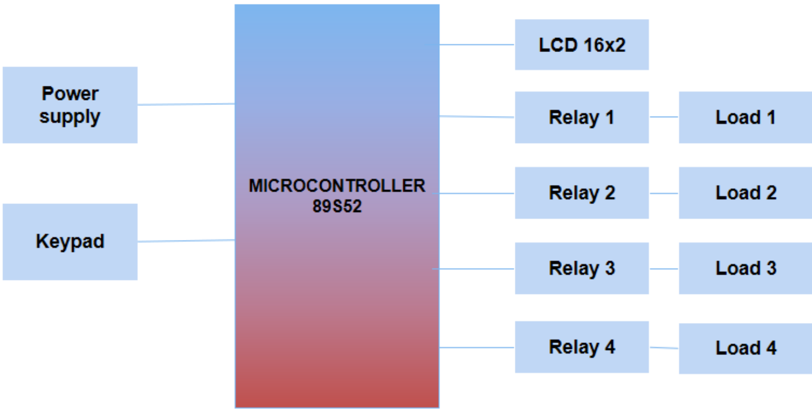

Figure 19 Block Diagram

Block diagram consists of Microcontroller AT89S52 from 8051 family, hex keypad, 19x2 LCD, SPDT relays and loads as feeder. First DC 5volts power supply is given to the system, then password is entered with the help of hex keypad that is interfaced with microcontroller, after that microcontroller reads the entered password, and compared the password with preinstalled password. Now SPDT relays sense the electric signal from microcontroller and relay will operate(break/close) the circuit. If the password is correct, then the loads will turn ON/OFF according to the entered password. If the entered password is incorrect, then the loads will remain in the same condition as before. LCD is connected to microcontroller to display the information.

3.2. Flow Chart:

Flow chart shows the steps in sequential order and workflow or process.

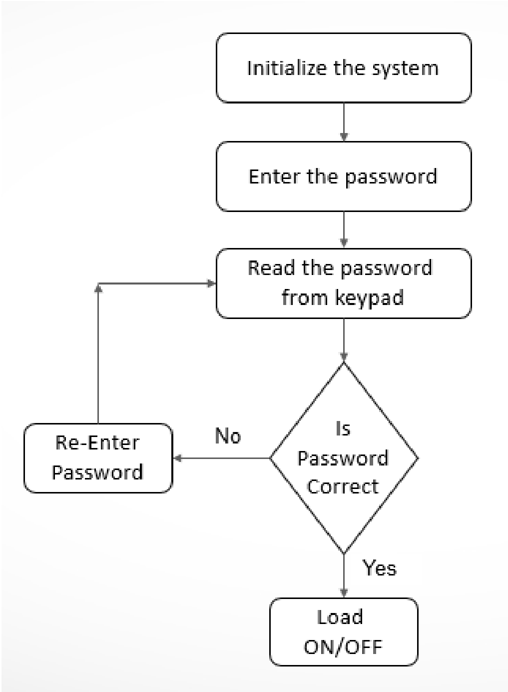

Figure 20 Flow Chart

Following are the steps or process to operate the prototype:

- In step 1, initialize the system that means power supply is given to the system.
- In step 2, After supply is given to the system, now enter the password with the hex keypad that is interfaced with microcontroller.
- In step 3, Entered password is read by microcontroller and display the information on LCD.
- In step 4, The password is checked, if password is not correct then re-enter the password and again microcontroller reads the password.
- If the password is correct, then step 5 will processed that is load will be turn ON/OFF according to the entered password.

3.3. Circuit Diagram:

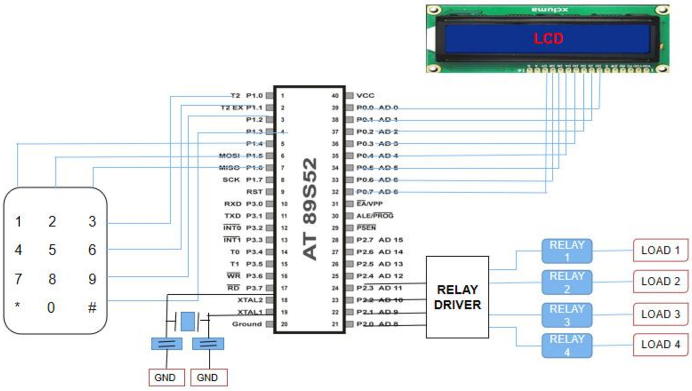

Figure 21 Circuit Diagram

The circuit diagram consists of microcontroller AT89S52 from 8051 family, hex keypad, 16x2 LCD, crystal oscillator, relay driver, Single pole double through(SPDT) relay. There is a total of 40pins in microcontroller 8051. Here we give power supply to microcontroller by using the pin 31&40. The Hex keypad is connected to microcontroller by using the pin 1 to 7 i.e., port 1. Crystal oscillator are connected to pin 18&19. LCD is connected to microcontroller by using pin 32 to 39 i.e., port 0. Relay driver is connected to pins 21 to 24 i.e., port 2. Relays are connected to the relay driver and loads are connected to relays. Pin 20 is connected to grounded, and Pin 9 is used for resetting the system.

CHAPTER 4

Hardware Details

4.1. Hardware Construction:

The AC power supply is given to the step-down transformer that is our first component in the whole circuit. The transformer in then to the diode and capacitor then from their power goes to regulator. It is connected to the power button that resets the circuit. Further the led starts glowing showing the indication of 5V supply. Then circuit is further connected to microcontroller. On pin 18 and 19 of microcontroller the crystal oscillator is connected that gives clock signal to microcontroller. The ceramic capacitors are connected with oscillator that acts as a filter. Then LCD, crystal oscillator, HEX keyboard and PNP and NPN transistors are connected to the microcontroller. Further relay is connected to transistors then relay is connected to the load. Contrast controller of LCD is connected on pin3 of LCD to control the contrast of the LCD.

4.2. Hardware Working:

In this project, the input power is divided at two sections. First one is by step-down transformer from 220V to 12V, the using diode and capacitor we converted 12V AC to DC and then at second section by using regulator we converted power from 12V to 5V DC and is given to microcontroller, LCD & other components and our system starts working. During maintenance maintainer may met with fatal accident. So, for protection of maintainer, relay is operated by password. This is done with the help of microcontroller. First of all the password is pre-set by programming. The password is entered using HEX keyboard. When we entered the password by the keypad if it is matched by pre-set password then the microcontroller sends a signal to trip the password-based relay. The relay will stop the power supply to the load. Again when maintenance is done, password to be enter and if it matched with pre-set password, signal is send by microcontroller and relay ON and it will supply power to load. In other cases if password is not matched with pre-set password then system will display wrong password at LCD display and will not operate the relay.

CHAPTER 5

Results and Discussion

The following steps and snapshots show the prototype working scenario and outcomes of this system.

5.1. When Password is Correct:

Figure 22 Display to show the enter password.

Figure 23 Entering the password.

Figure 24 Entered password is accepted.

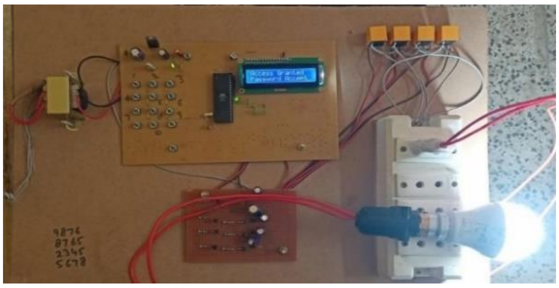

Figure 25 Load ON

First power supply is given to the system then enter the password with the help of hex keypad that is interfaced with microcontroller then microcontroller reads the entered password and compared the password with preinstalled password, now if the password is matched with preinstalled password, then load will turn ON/OFF. A 16x2 LCD is also connected to microcontroller to display the information.

5.2. When Password is Incorrect:

Figure 26 Display to show the enter password.

Figure 27 Entering the password.

Figure 28 Entered password is incorrect and not accepted.

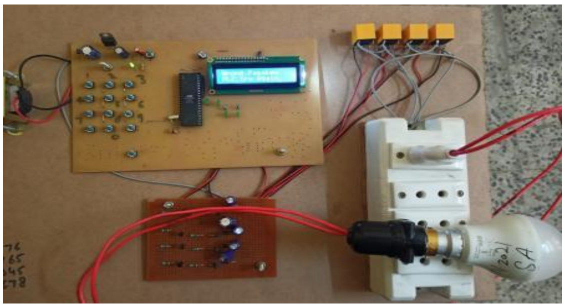

Figure 29 LOAD STILL REMAINS OFF.

First power supply is given to the system then enter the password with the help of hex keypad that is interfaced with microcontroller then microcontroller reads the entered password and compared the password with preinstalled password, now if the password is incorrect then the system will not operate and load still in same state as before.

5.3. Table:

Let us consider, initially all the loads is in Off state. Figure 30 represents all the possibilities of all the loads according to entered password, When the password is entered by hex keypad that is connected to the microcontroller then password is compared to preinstalled password, if the password is correct then system will operate. For example, if the password is correct for all the loads, then all loads will turn ON, if the password is correct for particular load, then that load will turn on.

When password is correct	LOAD A	LOAD B	LOAD C	LOAD D
For All Load	ON	ON	ON	ON
For Load A	ON	OFF	OFF	OFF
For Load B	OFF	ON	OFF	OFF
For Load C	OFF	OFF	ON	OFF
For Load D	OFF	OFF	OFF	ON
For Load A, B	ON	ON	OFF	OFF
For Load A, C	ON	OFF	ON	OFF
For Load A, D	ON	OFF	OFF	ON
For Load B, C	OFF	ON	ON	OFF
For Load B, D	OFF	ON	OFF	ON
For Load C, D	OFF	OFF	ON	ON
For Load A, B, C	ON	ON	ON	OFF
For Load B, C, D	OFF	ON	ON	ON
For Load A, C, D	ON	OFF	ON	ON

Figure 30 Table Represents all the Load ON/OFF Possibilities.

CHAPTER 6

Merits and Applications

6.1. Merits

- It improves the line man safety.
- Project is simple and easy.
- Uses commonly available components.
- Most useful to operate in the public areas.
- Save the life of line man.
- User friendly operation of main line.
- Easy to install and operate.
- Cost effective.
- Easy to maintain and repair.

6.2. Applications

- It can be used anywhere in the substation to trip the circuit.
- Most useful to operate in the public areas.
- This system is used in buildings and houses.
- Used in hotels and shopping malls to save the power.
- Used in electrical substations to ensure line man safety.

CHAPTER 7

Conclusion and Future Scope

7.1. Conclusion

- It can work on given known password.
- The password to operate can be changed and system and system can be operated efficiently with the changed password.
- It gives no scope of password stealing.
- It ensures the lineman safety.
- There is also a provision of changing the password by changing the code.

7.2. Future Scope

- Development in electrical power transmission system require the use of circuit breaker with increasing breaking capacity.
- It can also be interfaced with a GSM modem for remotely controlling the electronic circuit breaker via SMS.

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APPENDIX

A. Source Code:

```
#include<reg51.h>

#include<string.h>

#define lcdport P0


sbit col1=P1^4;

sbit col2=P1^5;

sbit col3=P1^6;

sbit col4=P1^7;

sbit row1=P1^0;

sbit row2=P1^1;

sbit row3=P1^2;

sbit row4=P1^3;


sbit rs=P2^0;

sbit en=P2^1;

sbit buzzer=P2^2;


sbit led=P2^7;

sbit led1=P2^6;

sbit led2=P2^5;

sbit led3=P2^4;

char pass[4],i=0,b_pressed=0,key_pressed=0;


char password1[4] ="9876";

char password2[4] ="8765";

char password3[4] ="2345";

char password4[4] ="5678";
```

```
sbit button1 = P3^4;

void delay(int itime)
{
   int i,j;
   for(i=0;i<itime;i++)
   for(j=0;j<1275;j++);
}

void daten()
{
   rs=1;
   en=1;
   delay(5);
   en=0;
}

void lcddata(unsigned char ch)
{
   lcdport=ch & 0xf0;
   daten();
   lcdport=(ch<<4) & 0xf0;
   daten();
}

void cmden(void)
{
   rs=0;
   en=1;
   delay(5);
```

```
    en=0;
}

void lcdcmd(unsigned char ch)
{
    lcdport=ch & 0xf0;
    cmden();
    lcdport=(ch<<4) & 0xf0;
    cmden();
}

void lcdstring(char *str)
{
    while(*str)
    {
        lcddata(*str);
        str++;
    }
}

void lcd_init(void)
{
    lcdcmd(0x02);
    lcdcmd(0x28);
    lcdcmd(0x0e);
    lcdcmd(0x01);
}

void keypad()
{
        int cursor=192,flag=0;
```

```
lcdcmd(1);
lcdstring("Enter Ur Passkey");
lcdcmd(0xc0);
i=0;
while(i<4)
{
flag=cursor;
 col1=0;
 col2=col3=col4=1;
 if(!row1)
 {
    lcddata('1');
    pass[i++]='1';
    cursor++;
    while(!row1);
      }

  else if(!row2)
 {
    lcddata('4');
    pass[i++]='4';
    cursor++;
    while(!row2);
    }

  else if(!row3)
 {
    lcddata('7');
    pass[i++]='7';
    cursor++;
    while(!row3);
```

```
  }
 else if(!row4)
{
  lcddata('*');
  pass[i++]='*';

  while(!row4);
  }

col2=0;
col1=col3=col4=1;
if(!row1)
{
  lcddata('2');
  pass[i++]='2';
  cursor++;
  while(!row1);
  }

 else if(!row2)
{
  lcddata('5');
  pass[i++]='5';
  cursor++;
  while(!row2);
  }

 else if(!row3)
{
  lcddata('8');
  pass[i++]='8';
```

```
   cursor++;
   while(!row3);
   }

 else if(!row4)
{
   lcddata('0');
   pass[i++]='0';
   cursor++;
   while(!row4);
   }

col3=0;
col1=col2=col4=1;
if(!row1)
{
   lcddata('3');
   pass[i++]='3';
   cursor++;
   while(!row1);
   }

 else if(!row2)
{
   lcddata('6');
   pass[i++]='6';
   cursor++;
   while(!row2);
   }

 else if(!row3)
```

```
{
  lcddata('9');
  pass[i++]='9';
  cursor++;
  while(!row3);
  }

 else if(!row4)
{
  lcddata('#');
  pass[i++]='#';
  cursor++;
  while(!row4);
  }

 col4=0;
col1=col3=col2=1;
if(!row1)
{

  while(!row1);
  }

 else if(!row2)
{
  lcddata('B');

  while(!row2);
  }

 else if(!row3)
```

```
   {
     lcddata('C');
     pass[i++]='C';
     cursor++;
     while(!row3);
     }

    else if(!row4)
   {
     lcddata('D');
     pass[i++]='D';
     cursor++;
     while(!row4);
       }
   if(i>0)
   {
      if(flag!=cursor)
      delay(100);
      lcdcmd(cursor-1);
      lcddata('*');
   }
 }
}

void accept()
{
                        lcdcmd(1);
                        lcdstring("Access Granted ");
   lcdcmd(192);
   lcdstring("Password Accept");
                        delay(300);
```

```
//                      key_pressed = 0;
  }

void wrong()
{
   buzzer=0;
   lcdcmd(1);
    lcdstring("Wrong Passkey");
   lcdcmd(192);
   lcdstring("PLZ Try Again");
   delay(1000);
   buzzer=1;
}
void read_sw()
{
            if((button1 == 0) &&(key_pressed == 0))
                                b_pressed = 1;
            else if((button1 == 1) &&(b_pressed == 1))
            {
                                key_pressed = 1;
                                b_pressed = 0;
            }
}
void main()
{
   buzzer=1;
   lcd_init();

   lcdstring("Smart LOAD SYS ");
   lcdcmd(0xc0);
             lcdstring("   Protected   ");
```

```
delay(400);
while(1)
{
 i=0;
                    read_sw();
                    if(key_pressed == 1)
                    {
                            keypad();
                            if(strncmp(pass,password1,4)==0)        //PASSWORD1
                            {
                                    if(led == 0)
                                            led = 1;
                                    else led = 0;
                                    accept();
                                    delay(100);
                            }
                            else if(strncmp(pass,password2,4)==0) //PASSWORD2
                            {
                                    if(led1 == 0)
                                            led1 = 1;
                                    else led1 = 0;
                                    accept();
                                    delay(100);
                            }
                            else if(strncmp(pass,password3,4)==0) //PASSWORD3
                            {
                                    if(led2 == 0)
                                            led2 = 1;
                                    else led2 = 0;
                                    accept();
                                    delay(100);
```

```
}
else if(strncmp(pass,password4,4)==0) //PASSWORD4
{
    if(led3 == 0)
        led3 = 1;
    else led3 = 0;
    accept();
    delay(100);
}
else
{
lcdcmd(1);
lcdstring("Access Denied");
wrong();
delay(100);
}
}
}
}
```